

Universidad Politécnica de Aguascalientes

Project Plan

Medical Information System — Centralized Cloud Migration

Case Study: Centro Médico del Bajío

Subject:	IT Project Management — Unit II
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Team:	Carlos Arturo Orozco Huitron — UP230562
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Duration:	8 weeks (40 business days)
Timeline:	June 1 – July 31, 2026
Date:	June 26, 2026

1. Activity List

The project is broken down into 10 sequential activities on the critical path. Each activity lists its ID, description, predecessor, duration and responsible party. The baseline duration is 38 business days, leaving 2 days of slack within the 8-week (40 business-day) window.

ID	Activity	Pred.	Dur.	Responsible
C1	Define cloud hosting environment and storage requirements		3 d	PM, Cloud Specialist
C2	Analyze medical records and document types from the 3 branches	C1	4 d	PM, Medical Staff
C3	Design cloud database and storage architecture	C2	4 d	Cloud Specialist
C4	Evaluate cloud hosting providers (criteria table) and acquire digitalization equipment	C3	5 d	PM, Procurement
C5	Install and configure equipment in the 3 branches	C4	3 d	Implementation Team
C6	Digitize patient clinical records	C5	5 d	Digitization Team
C7	Digitize clinical analyses and complementary documents	C6	5 d	Digitization Team
C8	Migrate and upload digitized records to the cloud platform	C7	4 d	Cloud Specialist, IT
C9	Analyze and remove duplicate records	C8	3 d	IT Team
C10	Integration and validation testing	C9	2 d	PM, IT Team

Table 1. Project activity list. Critical path: C1→C2→C3→C4→C5→C6→C7→C8→C9→C10 (38 business days).

Budget Allocation per Activity

ID	Activity	Budget (MXN)
C1	Define cloud environment	8,000
C2	Analyze records	10,000
C3	Design architecture	12,000
C4	Provider selection + equipment	55,000
C5	Install equipment	12,000
C6	Digitize patient records	15,000
C7	Digitize analyses	15,000
C8	Migrate to cloud	10,000
C9	Remove duplicates	6,900
C10	Testing	4,000
	TOTAL (BAC)	147,900

Table 2. Budget allocation per activity (BAC = \$147,900 MXN).

2. Gantt Chart

The Gantt chart below shows the schedule of the 10 activities along the critical path, with the Week-4 review checkpoint (June 26) marked and the two slippage events reported at that meeting: the two-day sick-leave delay on C1 and the three-day extension on C6. The management target line indicates the go-live date moved one week earlier.

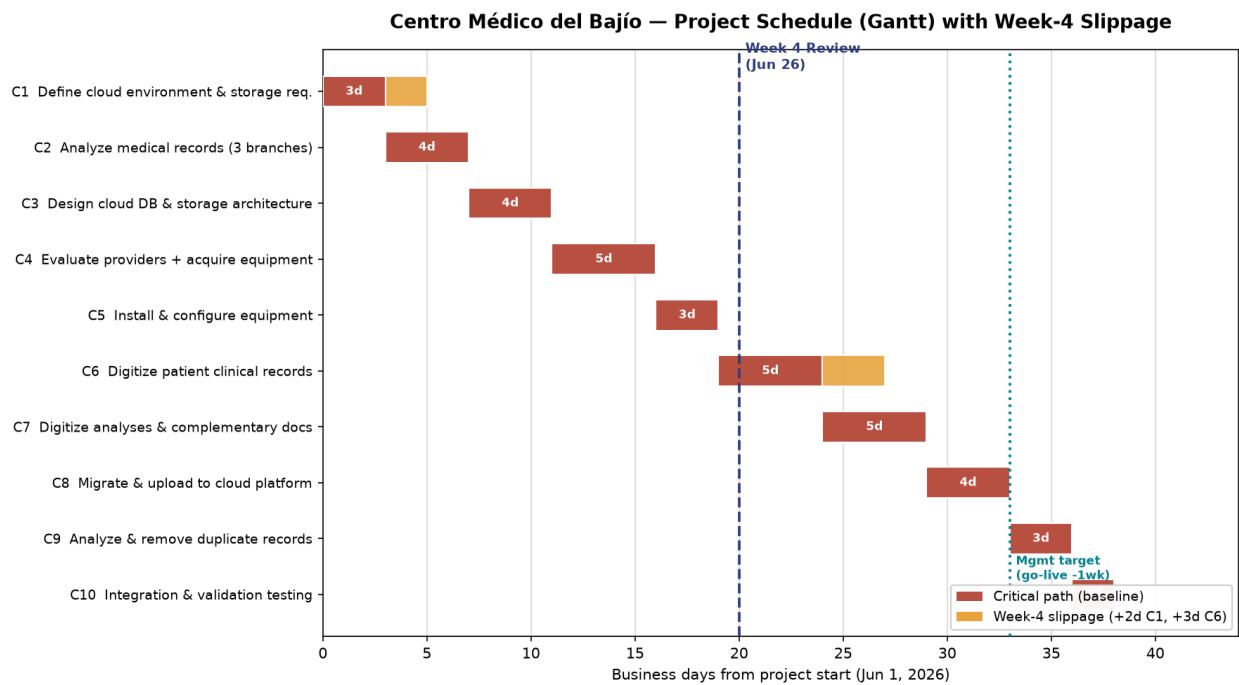


Figure 1. Project Gantt chart with Week-4 slippage overlay.

3. Project Closure Report

Hits (What Was Done Right)

- Successful deployment of the centralized cloud infrastructure (4 VMs, 32 GB RAM).
- 100% elimination of the 53,208 duplicated clinical records.
- Query response time achieved under 2.0 seconds across the 3 branches.
- The weighted criteria table gave an objective, defensible provider selection (Provider 1, 7.8/10).
- Change control absorbed the Week-4 disruptions without missing the July 31 go-live.

Failures (What Went Wrong)

- The initial activity (C1) was delayed 2 days by staff sick leave — no backup owner had been assigned.
- Three procurement deliverables were still pending at the Week-4 review.
- Activity C6 was underestimated — the patient-record volume required 3 extra days.
- The Earned Value formulas were initially recorded incorrectly (SV and CV swapped) and had to be corrected.

Improvements (For the Next Project)

- Assign a backup owner for every critical-path activity to absorb absences.
- Estimate digitization with a per-record parametric model instead of a flat duration.
- Pre-qualify vendors and lock delivery-time SLAs during Week 1.
- Standardize Earned Value templates so the SV and CV formulas cannot be swapped.
- Keep the short daily stand-up alignment meetings introduced during the rework.

Goals (Objective Achievement)

The strategic objective — consolidating the three branches onto a centralized cloud platform with 0% record redundancy, sub-2-second query response, within the \$147,900 MXN budget, and delivered by July 31, 2026 — was achieved at 100% compliance despite the disruptions reported at the Week-4 review. Corrective action through change control and the contingency reserve allowed the project to recover the schedule and meet all acceptance criteria.

4. Earned Value Analysis — Week 4 Summary

Earned Value Management (EVM) was the primary analytical technique used throughout project execution. At the Week-4 review (June 26, 2026, day 20), the project was assessed against the \$147,900 MXN Budget at Completion (BAC). Planned value (PV) of work scheduled by day 20 was \$100,000 MXN. Earned value (EV) of work actually completed was \$93,000 MXN (46% completion). Actual cost (AC) incurred was \$97,200 MXN.

Correct PMBOK Formulas

Schedule Variance: $SV = EV - PV$ (BCWP – BCWS) = \$93,000 – \$100,000 = **–\$7,000 MXN** (behind schedule).

Cost Variance: $CV = EV - AC$ (BCWP – ACWP) = \$93,000 – \$97,200 = **–\$4,200 MXN** (over budget).

Note: The two formulas were initially recorded swapped on the Week-4 whiteboard. The corrected formulas above follow the PMBOK standard.

Metric	Formula	Value	Interpretation
BAC	Total approved budget	\$147,900	Budget at Completion
PV (BCWS)	Planned value @ day 20	\$100,000	Budgeted cost of work scheduled
EV (BCWP)	Earned value @ day 20	\$93,000	Budgeted cost of work performed
AC (ACWP)	Actual cost @ day 20	\$97,200	Actual cost of work performed
SV	$EV - PV$	–\$7,000	Behind schedule
CV	$EV - AC$	–\$4,200	Over budget
SPI	EV / PV	0.93	< 1 → behind schedule
CPI	EV / AC	0.957	< 1 → over budget
EAC	BAC / CPI	\$154,545	Estimate at Completion
ETC	$EAC - AC$	\$57,345	Estimate to Complete
VAC	$BAC - EAC$	–\$17,418	Variance at Completion (within contingency)
TCPI	$(BAC - EV) / (BAC - AC)$	1.08	Efficiency needed to finish at BAC

Table 3. Earned Value Analysis — Week 4 (June 26, 2026).

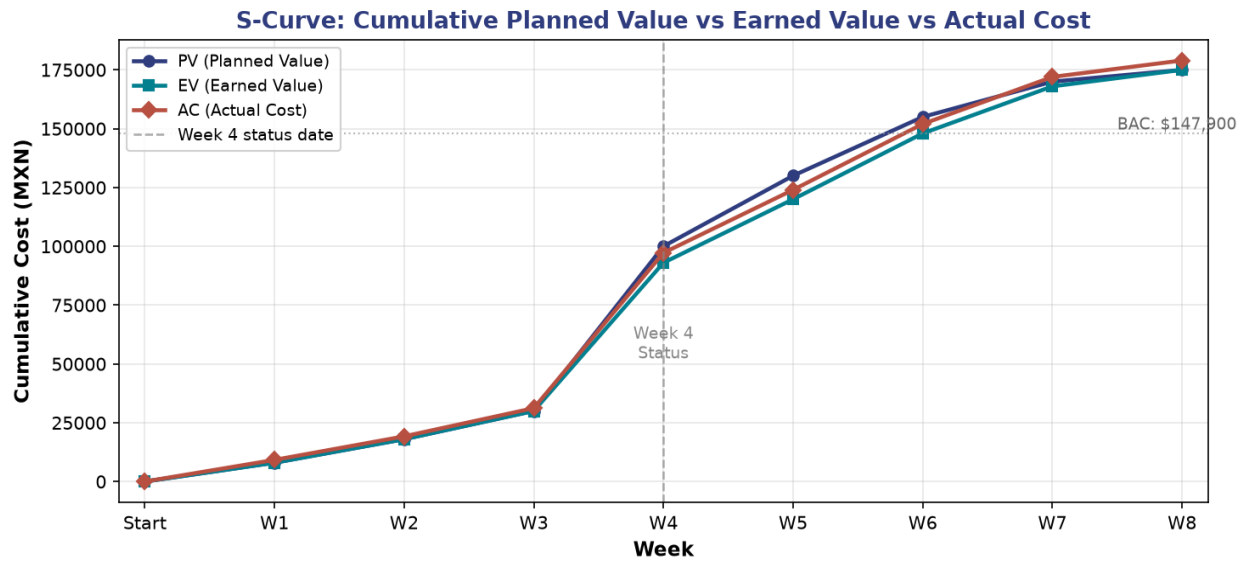


Figure 2. S-Curve — cumulative PV, EV, and AC over project timeline.

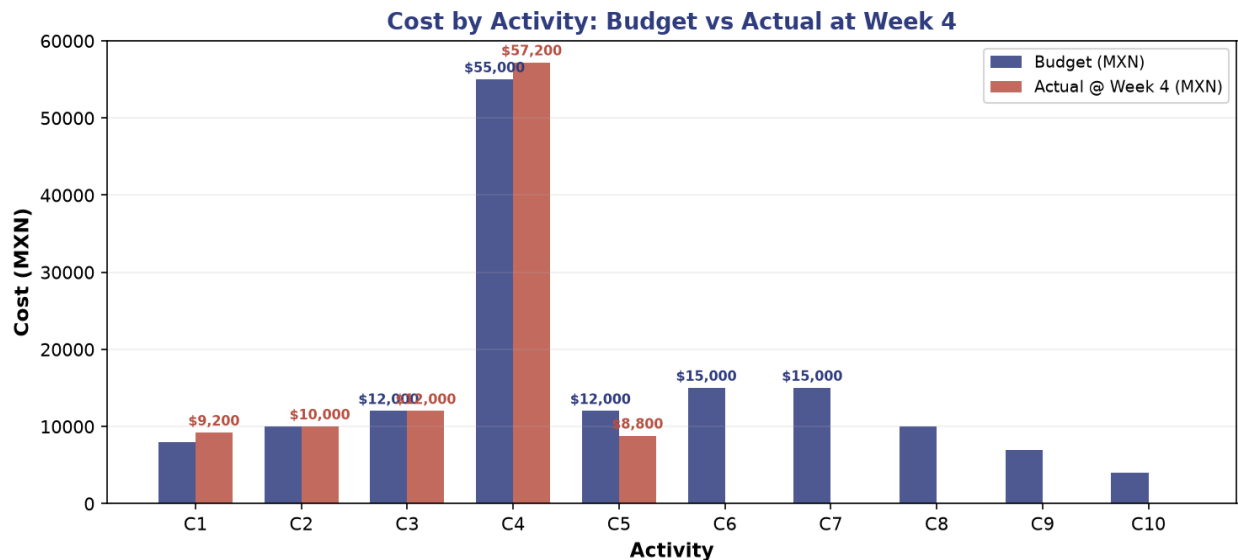


Figure 3. Cost by activity — budget vs actual at Week 4.

Interpretation: An SPI of 0.93 and CPI of 0.957 indicate the project was slightly behind schedule and over budget at the Week-4 checkpoint. The EAC of \$154,545 MXN represents a \$6,645 (4.5%) increase over the BAC, fully covered by the 10% contingency reserve of \$14,790 MXN. Recovery actions — crashing the digitization path (CR-004) and fast-tracking migration/deduplication — are projected to bring SPI back above 1.0 by Week 6.